

SiteGuard AI

Complete Database Design & Documentation

AI-Powered Construction Site Safety Monitoring System

1. Project & Database Overview

SiteGuard AI is a Java and AI-based construction safety monitoring system. Computer vision analyzes camera frames or uploaded media to identify potential safety issues such as missing helmets, missing safety vests, restricted-zone entry, fire/smoke and other unsafe conditions. The relational database stores projects, sites, cameras, AI detections, confirmed violations, alerts, users and audit history.

2. Technology Stack

Layer	Technology
Backend	Java 17/21, Spring Boot, Spring Security, Spring Data JPA / Hibernate
Database	MySQL 8.x
AI / Computer Vision	YOLO / ONNX Runtime, OpenCV
Frontend	React, Tailwind CSS
Security	JWT authentication, BCrypt password hashing
API	REST APIs; WebSocket can be used for real-time alerts

3. Database Objectives

- Maintain structured information about construction projects and monitored sites.
- Store camera configuration and raw AI detection results.
- Convert AI detections into rule-based safety violations.
- Track incident severity, acknowledgement and resolution.
- Generate alerts for safety officers and authorized users.
- Maintain an audit trail for important user actions.

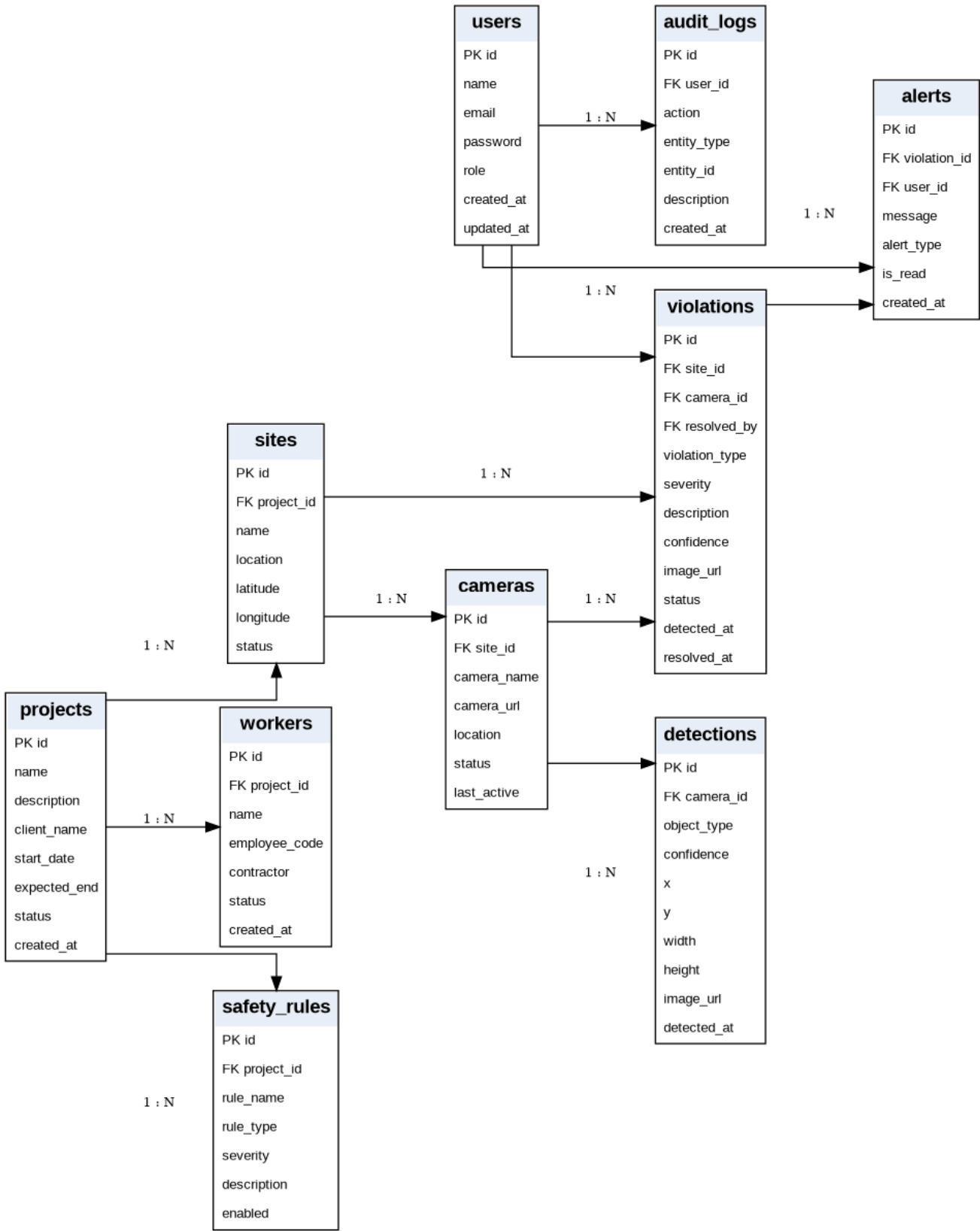
4. Database Tables at a Glance

Table	Purpose
users	Authentication, roles and system users
projects	Construction project master data
sites	Physical project sites/zones
cameras	CCTV camera configuration
workers	Workers assigned to projects
safety_rules	Configurable safety policies
detections	Raw AI/computer-vision detections
violations	Confirmed safety incidents
alerts	Notifications generated from incidents
audit_logs	Security and activity audit trail

5. Entity Relationship Diagram

The ER diagram shows the main entities and their relationships in the SiteGuard AI database.

1 : N



Key design principle: raw AI detections are stored separately from confirmed safety violations. This allows Spring Boot business rules to decide whether an AI detection should become an incident.

6. Database Relationships & Processing Flow

Relationship Summary

Parent	Child	Cardinality	Purpose
projects	sites	1 : N	Project contains construction sites/zones
sites	cameras	1 : N	Site can have multiple cameras
projects	workers	1 : N	Workers belong to a project
projects	safety_rules	1 : N	Rules are configured per project
cameras	detections	1 : N	Camera produces AI detections
sites	violations	1 : N	Site contains safety incidents
cameras	violations	1 : N	Camera can generate incidents
users	violations	1 : N	User can resolve incidents
violations	alerts	1 : N	Incident can generate alerts
users	alerts	1 : N	User can receive alerts
users	audit_logs	1 : N	User actions are audited

AI-to-Database Processing Flow

Step	Process
1	Camera / uploaded image provides a frame.
2	AI model detects objects and safety conditions.
3	Raw detection is stored in detections.
4	Spring Boot checks the active safety_rules for the project.
5	If a rule is violated, a record is created in violations.
6	High/Critical violations generate alerts.
7	Safety officer acknowledges or resolves the incident.
8	Important actions are recorded in audit_logs.

7. Database Design Notes

- Primary keys use BIGINT with auto-increment.
- Foreign-key columns are indexed by MySQL where applicable.
- Passwords are stored as BCrypt hashes, never as plain text.
- Evidence images/videos should be stored in file/object storage; MySQL stores their URL or path.
- Confidence values use DECIMAL(5,4), representing AI confidence from 0 to 1.
- ENUM values are used for controlled status/severity/role fields.

8. MySQL Table Structures

The following structures correspond to the MySQL DESC table_name output.

1. users

mysql> DESC users;

Field	Type	Null	Key	Default	Extra
id	BIGINT	NO	PRI	NULL	auto_increment
name	VARCHAR(100)	NO		NULL	
email	VARCHAR(150)	NO	UNI	NULL	
password	VARCHAR(255)	NO		NULL	
role	ENUM('ADMIN','SITE_MANAGER','SAFETY_OFFICER','VIEWER')	NO		NULL	
created_at	DATETIME	NO		NULL	
updated_at	DATETIME	NO		NULL	

7 rows in set

2. projects

mysql> DESC projects;

Field	Type	Null	Key	Default	Extra
id	BIGINT	NO	PRI	NULL	auto_increment
name	VARCHAR(150)	NO		NULL	
description	TEXT	YES		NULL	
client_name	VARCHAR(150)	YES		NULL	
start_date	DATE	YES		NULL	
expected_end	DATE	YES		NULL	
status	ENUM('PLANNED','ACTIVE','COMPLETED','ON_HOLD')	NO		NULL	
created_at	DATETIME	NO		NULL	

8 rows in set

3. sites

mysql> DESC sites;

Field	Type	Null	Key	Default	Extra
id	BIGINT	NO	PRI	NULL	auto_increment
project_id	BIGINT	NO	MUL	NULL	
name	VARCHAR(150)	NO		NULL	
location	VARCHAR(255)	YES		NULL	
latitude	DECIMAL(10,7)	YES		NULL	
longitude	DECIMAL(10,7)	YES		NULL	
status	ENUM('ACTIVE','INACTIVE')	NO		ACTIVE	

7 rows in set

4. cameras

mysql> DESC cameras;

Field	Type	Null	Key	Default	Extra
id	BIGINT	NO	PRI	NULL	auto_increment
site_id	BIGINT	NO	MUL	NULL	
camera_name	VARCHAR(100)	NO		NULL	
camera_url	VARCHAR(500)	YES		NULL	
location	VARCHAR(255)	YES		NULL	
status	ENUM('ACTIVE','INACTIVE','ERROR')	NO		ACTIVE	
last_active	DATETIME	YES		NULL	

7 rows in set

5. workers

mysql> DESC workers;

Field	Type	Null	Key	Default	Extra
id	BIGINT	NO	PRI	NULL	auto_increment
project_id	BIGINT	NO	MUL	NULL	
name	VARCHAR(120)	NO		NULL	
employee_code	VARCHAR(80)	YES	UNI	NULL	
contractor	VARCHAR(150)	YES		NULL	
status	ENUM('ACTIVE','INACTIVE')	NO		ACTIVE	
created_at	DATETIME	NO		NULL	

7 rows in set

6. safety_rules

mysql> DESC safety_rules;

Field	Type	Null	Key	Default	Extra
id	BIGINT	NO	PRI	NULL	auto_increment
project_id	BIGINT	NO	MUL	NULL	
rule_name	VARCHAR(150)	NO		NULL	
rule_type	VARCHAR(80)	NO		NULL	

Field	Type	Null	Key	Default	Extra
severity	ENUM('LOW','MEDIUM','HIGH','CRITICAL')	NO		NULL	
description	TEXT	YES		NULL	
enabled	TINYINT(1)	NO		1	

7 rows in set

7. detections

mysql> DESC detections;

Field	Type	Null	Key	Default	Extra
id	BIGINT	NO	PRI	NULL	auto_increment
camera_id	BIGINT	NO	MUL	NULL	
object_type	VARCHAR(80)	NO		NULL	
confidence	DECIMAL(5,4)	NO		NULL	
x	INT	YES		NULL	
y	INT	YES		NULL	
width	INT	YES		NULL	
height	INT	YES		NULL	
image_url	VARCHAR(500)	YES		NULL	
detected_at	DATETIME	NO		NULL	

10 rows in set

8. violations

mysql> DESC violations;

Field	Type	Null	Key	Default	Extra
id	BIGINT	NO	PRI	NULL	auto_increment
site_id	BIGINT	NO	MUL	NULL	
camera_id	BIGINT	NO	MUL	NULL	
resolved_by	BIGINT	YES	MUL	NULL	
violation_type	VARCHAR(100)	NO		NULL	
severity	ENUM('LOW','MEDIUM','HIGH','CRITICAL')	NO		NULL	
description	TEXT	YES		NULL	
confidence	DECIMAL(5,4)	YES		NULL	
image_url	VARCHAR(500)	YES		NULL	
status	ENUM('OPEN','ACKNOWLEDGED','RESOLVED','FALSE_POSITIVE')	NO		OPEN	
detected_at	DATETIME	NO		NULL	
resolved_at	DATETIME	YES		NULL	

12 rows in set

9. alerts

mysql> DESC alerts;

Field	Type	Null	Key	Default	Extra
id	BIGINT	NO	PRI	NULL	auto_increment
violation_id	BIGINT	NO	MUL	NULL	
user_id	BIGINT	NO	MUL	NULL	
message	TEXT	NO		NULL	
alert_type	VARCHAR(50)	NO		NULL	
is_read	TINYINT(1)	NO		0	
created_at	DATETIME	NO		NULL	

7 rows in set

10. audit_logs

mysql> DESC audit_logs;

Field	Type	Null	Key	Default	Extra
id	BIGINT	NO	PRI	NULL	auto_increment
user_id	BIGINT	NO	MUL	NULL	
action	VARCHAR(80)	NO		NULL	
entity_type	VARCHAR(80)	NO		NULL	
entity_id	BIGINT	YES		NULL	
description	TEXT	YES		NULL	
created_at	DATETIME	NO		NULL	

7 rows in set